

WANYUAN, China

WMO: 572370

Lat: 32.0744N

Lon: 108.0317E

Elev: 674

StdP: 93.49

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.2	-1.0	-11.1	1.6	4.1	-8.8	1.9	3.2	10.6	5.3	8.8	4.7	1.3	320	0.357

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	34.1	23.5	32.9	23.3	31.7	22.8	25.1	31.3	24.5	30.2	24.0	29.2	2.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.6	20.0	27.1	23.1	19.4	26.6	22.6	18.8	26.3	81.0	31.1	78.7	30.6	76.6	29.2	27.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
8.7	7.1	5.8	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-4.6	36.1	1.6	1.1	-5.8	36.8	-6.7	37.5	-7.6	38.1	-8.8	38.8
			-5.7	26.1	1.5	0.6	-6.8	26.6	-7.6	27.0	-8.5	27.3	-9.6	27.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	4.6	6.8	11.4	16.5	20.1	23.5	26.2	25.8	21.3	16.2	11.1	6.1	(6)
		DBStd	7.93	2.46	2.78	3.67	3.49	2.67	2.41	2.13	2.71	2.92	2.86	3.04	2.28	(7)
		HDD10.0	441	169	94	28	2	0	0	0	0	1	24	123		(8)
		HDD18.3	1735	427	322	217	73	14	1	0	6	78	218	380		(9)
		CDD10.0	2571	1	5	71	198	312	404	501	491	338	193	56	1	(10)
		CDD18.3	823	0	0	1	19	67	154	243	233	94	12	0	0	(11)
		CDH23.3	6813	0	0	23	208	497	1146	2179	2114	577	69	1	0	(12)
		CDH26.7	2623	0	0	3	48	134	403	912	926	188	8	0	0	(13)
(14)	Wind	WSAvg	1.8	1.9	2.1	2.2	2.2	2.0	1.6	1.5	1.6	1.6	1.6	1.7	1.8	(14)
(15)	Precipitation	PrecAvg	1236	4	10	37	89	136	143	248	191	227	105	39	8	(15)
(16)		PrecMax	2218	18	43	106	206	264	268	596	569	532	236	103	40	(16)
(17)		PrecMin	771	0	0	7	18	34	27	92	35	13	25	10	0	(17)
(18)		PrecStd	300	4	10	22	44	65	71	127	116	133	59	22	9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	15.2	18.3	26.2	30.6	32.0	34.1	35.2	35.8	33.6	27.9	22.0	15.3		(19)
(20)		MCWB	7.4	10.0	15.3	17.8	19.3	22.0	24.5	24.0	22.6	18.8	14.1	8.7		(20)
(21)		DB	12.3	15.5	22.8	28.0	29.8	32.2	33.8	34.4	30.9	25.4	19.7	13.2		(21)
(22)		MCWB	6.4	8.6	13.5	16.7	19.1	22.0	24.0	23.7	21.5	17.3	12.9	7.5		(22)
(23)		DB	10.4	13.6	20.4	25.7	28.0	30.6	32.7	33.1	28.8	23.2	17.7	11.6		(23)
(24)		MCWB	5.4	7.7	12.1	15.8	18.4	21.5	23.8	23.4	20.7	16.5	12.3	7.0		(24)
(25)		DB	8.9	11.6	18.0	23.4	26.2	29.1	31.4	31.7	26.6	21.1	15.8	10.1		(25)
(26)		MCWB	4.8	6.9	11.1	15.2	18.0	21.2	23.6	23.2	20.0	15.8	11.5	6.3		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	8.4	11.0	16.7	20.1	21.8	24.6	26.0	25.7	24.1	20.0	15.3	10.2		(27)
(28)		MCDB	12.9	15.9	23.2	27.7	27.1	30.2	32.1	32.7	30.6	25.0	19.4	13.3		(28)
(29)		WB	7.2	9.6	14.2	18.3	20.6	23.7	25.2	24.9	23.1	18.5	14.1	8.7		(29)
(30)		MCDB	10.9	13.7	20.5	24.8	26.1	28.7	31.1	31.6	27.9	22.5	17.8	11.7		(30)
(31)		WB	6.1	8.5	12.8	17.0	19.7	22.9	24.7	24.4	22.1	17.6	13.2	7.7		(31)
(32)		MCDB	9.7	12.3	19.0	23.4	25.1	27.9	30.2	30.5	26.0	21.3	16.3	10.6		(32)
(33)		WB	5.1	7.5	11.7	15.9	19.0	22.2	24.3	23.9	21.0	16.7	12.4	6.8		(33)
(34)		MCDB	8.4	11.2	17.1	21.7	24.3	26.8	29.3	29.5	24.9	20.2	15.0	9.4		(34)
(35)	Mean Daily Temperature Range	MDBR	6.9	7.4	9.1	9.9	9.4	8.8	8.8	9.3	8.0	7.5	7.5	7.0		(35)
(36)		5% DB	11.0	12.1	14.3	15.1	13.9	12.4	11.4	12.0	12.1	12.0	11.7	10.7		(36)
(37)		MCWBR	6.1	6.3	6.0	5.4	4.7	3.8	3.1	3.2	3.9	4.6	5.5	5.8		(37)
(38)		5% WB	9.7	9.8	12.7	12.7	10.7	9.5	9.6	10.3	9.5	9.4	9.2	8.8		(38)
(39)		MCWBR	5.7	5.5	5.6	5.3	4.3	3.5	3.0	3.1	3.7	4.1	4.7	5.1		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.611	0.665	0.678	0.649	0.599	0.584	0.589	0.610	0.599	0.606	0.626	0.586		(40)
(41)		taud	1.569	1.504	1.521	1.616	1.773	1.856	1.878	1.806	1.792	1.698	1.606	1.631		(41)
(42)		Ebn at Noon	570	593	634	684	726	734	728	705	688	633	555	562		(42)
(43)		Edh at Noon	224	263	276	260	224	206	200	212	207	212	214	201		(43)
(44)	All-Sky Solar Radiation	RadAvg	1.88	2.10	3.21	4.09	4.25	4.47	4.83	4.48	3.19	2.52	1.97	1.73		(44)
(45)		RadStd	0.32	0.42	0.32	0.50	0.57	0.50	0.73	0.56	0.45	0.46	0.35	0.31		(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47)	Regional (5 neighbors)	+0.35	N/A	N/A	+0.68	N/A	N/A	-38	-86	+97	+41	(47)

Nomenclature: See separate page